

Total No. of Questions : 9]

SEAT No. :

PE4238

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[6582]-9

S.E. (Civil)

ENGINEERING MATHEMATICS - III
(2019 Pattern) (Semester - III) (207001)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Question No. 1 is compulsory.
- 2) Answer Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7, Q.8 or Q.9.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) Attempt the following :

- a) If $\Sigma xy = 190, \bar{x} = 4, \bar{y} = 4, n = 10, \sigma_x = 1.732, \sigma_y = 2$ then correlation coefficient $r(x, y)$ is [2]
- i) 0.91287 ii) 0.8660
iii) 0.7548 iv) 0.5324
- b) The mean and standard deviation of binomial probability distribution are 36 and 3 respectively. Number of trials n is [2]
- i) 42 ii) 36
iii) 48 iv) 24
- c) The curl of vector field $\vec{F} = x^2y\vec{j} + xyz\vec{j} + z^2y\vec{k}$ at point $(0, 1, 2)$ is [2]
- i) $4\vec{i} - 2\vec{j} + 2\vec{k}$ ii) $4\vec{i} + 2\vec{j} + 2\vec{k}$
iii) $4\vec{i} + 2\vec{k}$ iv) $2\vec{i} + 4\vec{k}$
- d) ∇r^n is _____. [2]
- i) nr^{n-1} ii) $\frac{r^{n+1}}{n+1}\vec{r}$
iii) $\frac{3r^{n-2}}{r}$ iv) $n r^{n-2}\vec{r}$

P.T.O.

- e) $\nabla \times \vec{r}$ is [1]
- i) $\vec{r}$ ii) 3
- iii) $\frac{\vec{r}}{r}$ iv) $\vec{0}$

- f) Line of regression y on x is [1]

i) $y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$ ii) $y + \bar{y} = r \frac{\sigma_y}{\sigma_x} (x + \bar{x})$

iii) $x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$ iv) $x - \bar{x} = r \frac{\sigma_y}{\sigma_x} (y - \bar{y})$

- Q2) a) The first four moments of a distribution about the value 2 are 2, 10, 20 and 25. Find first four moments about mean, coefficient of skewness and kurtosis. [5]

- b) Calculate the coefficient of correlation for the following data. [5]

x	-1	1	2	4	6	7
y	-1	2	3	3	5	8

- c) An unbiased coin is thrown 10 times. Find probability of getting. [5]
- i) Exactly 6 heads.
- ii) At least 6 heads

OR

- Q3) a) Obtain regression line y on x for the following data [5]

x	6	2	10	4	8
y	9	11	5	8	7

- b) On an average 180 cars per hour pass a specified point on a particular road. Using poisson distribution. Find the probability that at least two cars pass the point in any one minute. [5]
- c) In a sample of 1000 cases, the mean score in a certain examination is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find the expected number of students scoring between 12 and 15 (both inclusive). [5]

(Given : $Z_1 = 0.4, A_1 = 0.1554$; $Z_2 = 0.8, A_2 = 0.2881$)

Q4) a) Find the angle between the tangent to the curve [5]

$$\vec{r} = (t^3 + 2)\vec{i} + (4t - 5)\vec{j} + (2t^2 - 6t)\vec{k} \text{ at } t = 0 \text{ and } t = 1.$$

b) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at point $(2, -1, 2)$ in the direction $2\vec{i} - 3\vec{j} + 6\vec{k}$. [5]

c) Given $\vec{u} = xyz\vec{i} + (2x^2z - y^2x)\vec{j} + xz^3\vec{k}$ [5]

$$\phi = xy + yz + z^2 \text{ at point } (1, 0, -1)$$

Find

i) $\nabla \cdot \vec{u}$

ii) $\nabla \times \vec{u}$

iii) $\nabla \cdot (\phi \vec{u})$

OR

Q5) a) If $\vec{r}(t) = t^2\vec{i} + t\vec{j} - 2t^3\vec{k}$ then evaluate $\int_0^1 \vec{r} \times \frac{d^2\vec{r}}{dt^2} dt$. [5]

b) Show that vector field $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ is irrotational. Find the scalar potential ϕ such that $\vec{F} = \nabla\phi$. [5]

c) For constant vector $\vec{a}$. Show that [5]

i) $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$

ii) $\nabla \times (\vec{a} \times \vec{r}) = 2\vec{a}$

- Q6) a)** Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ around the parabolic arc $y^2 = x$ joining points $(0, 0)$ and $(1, 1)$. [5]
- b)** Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ where 's' is the surface of cylinder $x^2 + y^2 = 4$, $z = 0$, $z = 3$ and $\vec{F} = 4xi - 2y^2j + z^2k$. [5]
- c)** Evaluate $\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, ds$ for $\vec{F} = yi + 3j + xk$ where 's' is the surface of the paraboloid $z = 1 - x^2 - y^2$, $z \geq 0$. [5]

OR

- Q7) a)** Using Green's theorem evaluate $\int_C \cos y \, dx + x(1 - \sin y) \, dy$ where 'c' is the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, $z = 0$. [5]
- b)** Find the workdone in moving a particle once round the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$, $z = 0$ under the force field given by [5]
- $$\vec{F} = (2x - y + z)i + (x + y - z^2)j + (3x - 2y + 4z)k$$
- c)** Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ where $\vec{F} = x^3i + y^3j + z^3k$ and 'S' is the surface of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant. [5]

Q8) a) Solve $\frac{\partial^2 y}{\partial t^2} = C^2 \frac{\partial^2 y}{\partial x^2}$ with boundary conditions [8]

i) $y(0, t) = 0$

ii) $y(l, t) = 0$

iii) $\left[\frac{\partial y}{\partial t} \right]_{t=0} = 0$

iv) $y(x, 0) = k(lx - x^2); 0 \leq x \leq l$

b) Solve $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, $u(x, t)$ satisfies the conditions [7]

i) u is finite $\forall t$

ii) $u(0, t) = 0$

iii) $u(l, t) = 0$

iv) $u(x, 0) = u_0$ for $0 < x < l$

OR

Q9) a) Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to conditions [8]

i) u is finite for all 't'

ii) $u(0, t) = 0$

iii) $u(l, t) = 0$

iv) $u(x, t) = \frac{3x}{l}$ for $0 \leq x \leq l$

b) Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the conditions [7]

i) $u(x, \infty) = 0, y \rightarrow \infty$

ii) $u(0, y) = 0$

iii) $u(10, y) = 0$

iv) $u(x, 0) = 100 \sin \frac{\pi x}{10}; 0 \leq x \leq 10$